

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

**Duration : 1 Hour
Total Marks:30**

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I G. GOWTHAMI / (192524285 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. GOWTHAMI

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- i. D1 cannot work Night shift
- ii. D2 must work before D3 (Morning < Afternoon < Night)
- iii. Only one doctor per shift
- iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles
- iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

- b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- c) Demonstrate how the robot applies AI concepts such as:
- Intelligent agents
 - Rationality
 - Environment type
- d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

ANSWERS

1. Constraint-Based Problem Solving (Backtracking Search)

Given Constraints

D1 cannot work Night.

D2 must work before D3 (Morning < Afternoon < Night).

Only one doctor per shift.

D3 cannot work Morning.

Each doctor must be assigned exactly one shift.

Step 1: Assign D2

Possible shifts: Morning, Afternoon, Night

Choose: D2 → Morning

Step 2: Assign D3

D3 cannot work Morning. Since D2 must work before D3, D3 can be Afternoon or Night.

Choose: D3 → Afternoon

Step 3: Assign D1

Remaining shift is Night.

But D1 cannot work Night.

Backtrack

Try another assignment

Assign:

D2 → Morning

D3 → Night

D1 → Afternoon

Check constraints:

✓ D1 not Night

✓ D2 before D3

✓ D3 not Morning

✓ One doctor per shift

✓ Every doctor assigned once

Final Assignment

Doctor

Shift

D1

Afternoon

D2

Morning

D3

Night

2. Robot Path Finding (BFS Search)

Assume the 5×5 grid:

```
S . . X .  
. X . X .  
. X . . .  
. . X X .  
X . . . G
```

Where:

S = Start

G = Goal

X = Obstacle

BFS Node Expansion

Open Queue

[S]

Expand S

[(1,0),(0,1)]

Expand (1,0)

[(0,1),(2,0)]

Expand (0,1)

[(2,0),(0,2)]

Expand remaining nodes until Goal reached.

Optimal Path

S

↓

↓

→

→

↓

↓

→

→

G

Path Cost

Each move = 1

Total moves = 8

Total Cost = 8

3(a). Constraint Analysis

1. Limited Movement

Robot moves only:

Up

Down

Left

Right

Effect: Robot cannot move diagonally.

2. Obstacles

Blocked cells cannot be crossed.

Effect: Robot searches for another path.

3. Limited Sensing

Robot sees only nearby cells.

Effect: Cannot plan the whole route initially.

4. Movement Cost

Normal move = 1

Risky cell = 3

Effect: Robot prefers safer paths.

5. Dynamic Environment

Robot updates knowledge while moving.

Effect: If new obstacles appear, it replans the path.

3(b). Solution Approach (Uniform Cost Search)

Step 1

Start from S.

Step 2

Insert S into the priority queue.

Step 3

Expand the node with the smallest path cost.

Step 4

Generate neighboring cells.

Step 5

Ignore blocked cells.

Step 6

Calculate cumulative cost.

Step 7

Update priority queue.

Step 8

Repeat until Goal (G) is reached.

Step 9

Return the least-cost path.

3(c). AI Concepts

Intelligent Agent

The robot senses its surroundings and performs actions to reach the goal.

Rationality

The robot selects the safest and lowest-cost path.

Environment Type

Partially Observable

Dynamic

Sequential

Discrete

Unknown

3(d). Justification

Uniform Cost Search is suitable because:

It always finds the minimum-cost path.

It handles different movement costs.

It avoids risky cells when possible.

It can update decisions when the environment changes.

It guarantees the optimal solution if all costs are positive.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly

Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided